

Master Math for JEE Main & JEE Advanced

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JEE Mains 2020 Jan Chapter wise Question Bank

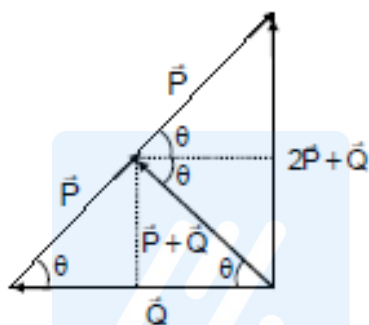
Mathematics in Physics

Q1

Magnitude of resultant of two vectors $\vec{P}$ and $\vec{Q}$ is equal to magnitude of $\vec{P}$. Find the angle between $\vec{Q}$ and resultant of $2\vec{P}$ and $\vec{Q}$.

7th Jan Evening

Sol

 90° 

So angle between $(2\vec{P} + \vec{Q})$ and $\vec{Q}$ is 90°

अतः $(2\vec{P} + \vec{Q})$ तथा $\vec{Q}$ के मध्य कोण 90° है।

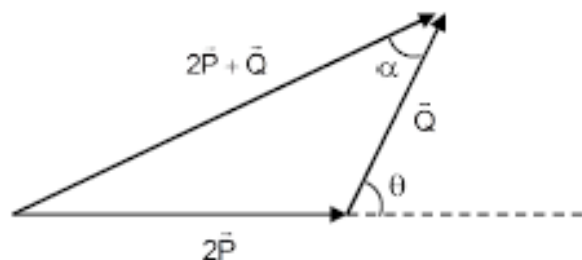
Alternate solution वैकल्पिक हल

$$|\vec{P} + \vec{Q}| = |\vec{P}|$$

$$P^2 + Q^2 + 2PQ\cos\theta = P^2$$

$$\Rightarrow Q + 2P\cos\theta = 0$$

$$\Rightarrow \cos\theta = -\frac{Q}{2P}$$



$$\tan \alpha = \frac{2P \sin \theta}{2P \cos \theta + Q} = \infty \quad \because [2P \cos \theta + Q = 0]$$

$$\alpha = 90^\circ$$

Given $\vec{p} = -\hat{i} - 3\hat{j} + 2\hat{k}$ and $\vec{r} = \hat{i} + 3\hat{j} + 5\hat{k}$. Find vector parallel to electric field at position $\vec{r}$

[Note that $\vec{p} \cdot \vec{r} = 0$]

दिया है द्विध्रुव $\vec{p} = -\hat{i} - 3\hat{j} + 2\hat{k}$ और $\vec{r} = \hat{i} + 3\hat{j} + 5\hat{k}$ तब स्थिति $\vec{r}$ पर विद्युत क्षेत्र के समान्तर सदिश ज्ञात करो।

[दिया है $\vec{p} \cdot \vec{r} = 0$]

(1) $\hat{i} + 3\hat{j} - 2\hat{k}$

(2) $3\hat{i} + \hat{j} + 2\hat{k}$

(3) $-3\hat{i} - \hat{j} - 2\hat{k}$

(4) $-\hat{i} + 3\hat{j} + 2\hat{k}$

9th Jan Morning

Sol

(1)

Since $\vec{p} \cdot \vec{r} = 0$

$\vec{E}$ must be antiparallel to $\vec{p}$

$\vec{E}$, $\vec{p}$ के प्रतिसमान्तर होगा

So, $\vec{E} = -\lambda(\vec{p})$

where λ is a arbitrary positive constant

जहाँ λ कोई एक धनात्मक नियतांक है

Now $\vec{A} = a\hat{i} + b\hat{j} + c\hat{k}$

अब $\vec{A} = a\hat{i} + b\hat{j} + c\hat{k}$

$\vec{A} \parallel \vec{E}$

$$\frac{a}{\lambda} = \frac{b}{3\lambda} = \frac{c}{-2\lambda} = k$$

so इसलिए $\vec{A} = \lambda k(\hat{i} + 3\hat{j} - 2\hat{k})$

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